The Brain's Conductor:
The Psychology of Focus, Flexibility, and Control

Location	William James Hall, B6	**Instructor**	Haley Keglovits, Ph.D.
Time	Tues/Thurs 12:00–1:15 PM	**Email**	hkeglovits@g.harvard.edu
Office Hours	TBD, WJH 7th Floor	**Canvas**	Link

Welcome to PSY1351!

Course Description: *Cognitive control, also known as executive function, is our ability to behave flexibly and not simply respond automatically to stimuli in our environment. It explains why you're able to avoid answering your phone while driving or concentrate on finishing your homework in a crowded cafe, but also why you might catch yourself walking to last semester's classes when the term changes, or find yourself opening social media apps on your phone without even realizing it. In this course we will dive into important topics in the study of control, discussing multitasking and task-switching; how we stop automatic actions and habits; how attention works; the ways that cognitive control changes over the lifespan, and how it is impacted by illness; whether brain training games make you smarter; how control interacts with short- and long-term memory; what role these processes plays in COVID-19 brain fog; and much more. Classes will involve lectures, demonstrations, activities, and discussion.*

Learning Objectives:

- Describe the scope of abilities and behaviors that are impacted by cognitive control
- Produce examples of automatic and controlled behaviors in their daily lives
- Compare and contrast theories of cognitive control with supporting empirical evidence
- Discuss the impact cognitive control deficits have on major psychiatric disorders
- Identify which brain regions are implicated in controlled processes
- Interpret data from common standardized tests of cognitive control
- Generate testable and falsifiable hypotheses about human behavior
- Integrate data across studies to produce predictions about unstudied populations

Course Assignments

Scientific Skills Papers (10 papers, 50% total): A weekly written assignment to build your skills reading, contextualizing, and assessing cognitive control research. Prompts and requirements will differ but all will be around 3-5 pages and due at 11:59 on Sundays.

Reading Comprehension Checks (10%): Brief (one question) in-class assessments of the readings assigned for the day, given at the **start** of each class. Your best 20 out of 24 total will be graded; no make-ups available.

Exit Tickets (10%): Every week, you'll respond to a set of rotating prompts to reflect on what we've learned, as a part of a group Canvas discussion. You must make at least two 100+ word posts per week, one by Friday at 11:59 PM (an original) and one by Sunday at 11:59 PM (an original or a follow up to a peer). Your best 10 weeks out of 12 will be graded (no week 1 or Thanksgiving).

Proposal (5%) + Final Paper (15%): An 8-10 page research paper developing a "new" study on a topic of your choice to add to our understanding of cognitive control. You'll explain your study's background; your variables, measurements, and analyses; your literature-grounded hypothesis; and what your (hypothetical) findings would mean in context. I will host individual meetings to discuss your ideas before the formal proposal is due.

Participation (5%) Participation means you should be prepared and willing to share your thoughts, questions, or reactions during class *regularly*. You will receive a check-in halfway through the semester; I take improvements into account when assigning a final grade. See classroom environment below for more details.

Attendance (5%) Regular attendance is essential for your success in this course. You must attend at least 75% of lectures (20 out of 26) to pass. Beyond that, you have two no-questions-asked excused absences. Any additional absences result in a 1% drop in your overall grade; up to 4 total. You will receive 1% credit for attending a final proposal one on one meeting with me. Absences do not require documentation (see below on extension fairness!).

More information about all assignments, and all deadlines, available on Canvas.

Course Schedule

Date	Topic	Readings	Assignments (Due 11:59)
Part 1: Introduction to Cognitive Control			
Topics: measuring control historically and now, the building blocks of control, and the investigative tools we use.			
Sept 3	Introduction to Control	Badre (2022), Chapter 1	
Sept 8	History of Studying Control	Burgess et al. (1998)	Society + Psychology Paper Due Sept. 13th
Sept 10	Measuring with Behavior	Milner (1963) and Bugg et al. (2008)	
Sept 15	Working Memory	Miller (1956) and Luck & Vogel (1997)	Background Paper Due Sept. 20th
Sept 17	Effort Allocation	Kool et al. (2010)	
Sept 22	Stopping and Inhibition	Logan et al. (1997) and Aron et al. (2003)	Pop Science Paper Due Sept. 27th
Sept 24	Error Monitoring	Notebaert et al. (2009) and Jentzsch & Dudschig (2009)	
Sept 29	Task Switching	Rogers & Monsell (1995)	Methods Paper Due Oct. 4th
Oct 1	Multi-Tasking	Pashler (1984) and Ophir et al. (2009)	
Oct 6	Stability-Flexibility Tradeoff	Fröber & Dreisbach (2017)	Control Failures Paper 1 Due Oct. 11th
Oct 8	Brain Measurements	Gehring et al. (1993) and Banich et al. (2000)	
Oct 13	Brain Regions	Bunge et al. (2003) and Botvinick et al. (1999)	Analysis Paper Due Oct. 18th
Oct 15	Why model?	Miller & Cohen (2001)	
Oct 20	Theories/Models of Control	Shenhav et al. (2013)	Control Failures Paper 2 Due Oct. 25th
Oct 22	Theories/Models of Control 2	Braver (2012) and Eisenreich et al. (2017)	
Part 2: Variations and Interactions of Cognitive Control			
Topics: lifespan changes, clinical populations, the influences of life experiences, and interactions with other cognition.			
Oct 27	Development: Early Life	Gerstadt et al. (1994) and Bunge et al. (2002)	Discussion Paper Due Nov. 1st
Oct 29	Development: Aging	Bugg et al. (2007)	
Nov 3	Mood and Thought Disorders	Adams et al. (2011) and Meiran et al. (2011)	New Findings Paper Due Nov. 8th
Nov 5	Psychosis	Cohen et al. (1999)	
Nov 10	Neurodegenerative Diseases	Muslimovic et al. (2005) and Stopford et al. (2012)	Next Steps Paper Due Nov. 15th
Nov 12	Life Factors	Tun et al. (2013) and Shannon et al. (2025)	
Nov 17	Injury and Illness	Ellemberg et al. (2007) and Duindam et al. (2022)	*Research Paper 1:1s Scheduled this week*
Nov 19	Can Control be Improved?	Melby-Lervåg & Hulme (2013)	
Nov 24	Research Proposal Workshop	*Guide to Writing the Psychology Paper*	**Final Paper Outline Due Nov. 25th at Noon**
Nov 26	Thanksgiving, No class		
Dec 1	Memory and Learning	Rmus et al. (2021) and Anderson et al. (2004)	**Final Due Dec. 13th**
Dec 3	Wrap Up	Bunge (2024)	

Grade Breakdown

	B+: 87–89	C+: 77–79	D+: 67–69	
A: 93–100	B: 83–86	C: 73–76	D: 63–66	E: <60
A-: 90–92	B-: 80–82	C-: 70–72	D-: 60–62	

Course Policies

Office Hours

Office hours are designated times that I set aside to engage with your questions or thoughts that may come up during class or on assignments. You are welcome to come talk to me about the course, how to write well in psychology, or even what it is like to do a Ph.D. They are even a great place to come do your work – you are very welcome to sit and read/write and ask me or classmates present clarifying questions as you work. My previous students have liked this option as a set time in their week to work on class assignments.

The default format is a drop-in basis (meaning you do not have to stay for the whole hour, you can come and go as you please, and you do not need to tell me ahead of time that you will come) and are for your benefit. Because of this open-door nature, please shoot me an email if you would like to discuss something without others present so we can schedule an uninterrupted time slot. Occasionally, I'll host themed sessions (ex: how to read scientific papers at the start of the semester). Let me know if there are other topics you'd find helpful for a more structured discussion.

Academic Integrity and Generative AI

Plagiarism is implicitly or explicitly taking credit for something that is not yours. In this course, you may collaborate with humans or tools only in the ways outlined below, and only with proper documentation. Submitting some(one/thing) else's work as your own not only risks academic consequences (including referral to the Honor Council), but it also cheats you out of the education you're investing your time and energy in. Harvard's Honor Code || FAS Handbook for Students

Classmate Collaboration: One of the great things about being in a university setting is that you get to learn through discussion and teamwork. Sharing ideas can be a valuable part of the learning process. I encourage you to talk with your classmates when you're reading papers or thinking about what you want to write about. I will include a place on paper submissions for you to note who (if anyone) you talked about the readings / your assignments with. This is not a trick! I genuinely encourage you to talk these papers through before writing them (at which point, all work must be individual). Office hours are a great place to collaborate with your peers.

Campus Resources: The Writing Center is an excellent resource to support your writing skills specifically, through tutors and drop-in hours. They are specifically trained in college writing, and help students at all stages. Psychology even has a designated department fellow who can help you with the specifics of psychology writing. The Academic Resource Center provides more general academic coaching which may be helpful as you manage many readings and assignments across your courses. I highly recommend use of these resources, and they do not require documentation.

Generative AI: You may use AI as a collaborator in the writing process to compose up to but not more than 25% of your essay assignments. However, any AI-generated text or ideas must be highlighted in blue font. The 25% limit refers not only to the literal word count, but also to the overall thought process that goes into the paper – it must be primarily your own work. Assignments that contain AI generated material but lack this clear visual distinction will be considered plagiarism. If AI is used during the revision process to improve grammar and clarity **only** (it was not used to generate the sentence or its ideas) this use does not count toward the limit and does not need to be highlighted in blue font. However, you must include a note with your references documenting this. We will discuss this policy more during the first class session.

You may use AI to assist in reading assigned papers, but you do so at your own risk and are responsible for an accurate understanding of the research. You may not use AI for the exit tickets – these do not need to be formal language, and I am genuinely asking for your thoughts.
No matter what, you take responsibility for all work you turn in, even if you collaborated on its generation. If you're unsure about what counts as appropriate collaboration or how to cite something, please ask during office hours.

Accommodations and Support

If you have academic accommodations through the Disability Access Office (DAO), please share your letter with me by the end of the second week of the term so we can make sure you get support. If you do not have formal accommodations yet but have concerns about completing assignments or staying on track in this class, I encourage you to reach out to me so we can discuss campus resource options. I'm always happy to talk during regular office hours or by scheduling a one on one, and our conversations will remain private. In some cases, I may need to consult with DAO to make sure we're applying accommodations appropriately or connecting you with the right resources. My goal is always to facilitate your learning. If something changes during the semester, please come talk to me as soon as possible.

If anything unexpected comes up that might affect your ability to succeed in this course, please reach out. Don't worry that you're "bothering" me – I'm here to support you, including helping you navigate challenges.

Late/Dropped Assignments

I know things happen in life that interfere with your academics, which is why I build in dropped assignments (exit tickets and reading comprehension checks). In addition to these, you will also be allowed two no-questions-asked 24-hour passes to extend a paper deadline without incurring late penalties. You may combine these passes for a single paper or use them individually, but you cannot break them into pieces (i.e., you can't put 4 hours on an assignment) or use them on the final.

This does not require any documentation on your part and will be calculated automatically (but you must track your own). Any further late assignments will incur a 10% penalty for each additional 24 hours delayed. This policy is meant to account for any type of lateness, so take heed: even one minute late counts as a delay of a day. I recommend not waiting until the last minute to submit because none of us can predict technology issues!

My no documentation policy exists because requiring people to request an exception, rather than receiving one automatically, can be inherently unfair. Certain demographic groups, such as students experiencing mental health or other stigmatized challenges, are often less likely to ask for exceptions to class policies (PMID: 34725153, 18378834, 31008459).

If you're dealing with a situation that has caused you to miss multiple assignment deadlines or class sessions, that probably means that it is something that should be approached in coordination with your Resident Dean and/or the Disability Access Office. I'm happy to help you with those processes; please just let me know by email or during office hours.

Classroom Environment

Active Engagement

Our discussions will be richer when everyone's voice is part of the conversation. During our first class, we'll co-create expectations for what thoughtful, respectful discussion looks like. If speaking in class is a skill you are still developing, please come talk with me early in the semester. We can come up with a plan that helps you participate in a way that supports your learning. Just like thinking critically or writing clearly, finding and using your voice is a skill you'll carry beyond this course. I want to create a space where you feel safe practicing it.

Respectful Discourse

My goal is to create an inclusive environment where everyone feels valued, regardless of their background in psychology or personal characteristics. This includes actively listening, ensuring widespread participation, and treating your peers as learners (just like you!) with the same right to contribute and grow. If you have concerns or suggestions about inclusivity in this course, I hope that you will feel comfortable bringing your ideas to me so that we can work together towards improvement.

In this course, we will discuss the impact psychiatric conditions have on people's cognition. I ask that you work with me to do so with respect for all cognitive abilities: these conditions are heterogeneous, and so are the people living with them. You may or may not have personal or social experience with these conditions, but it is likely your peers do – the National Institute of Mental Health estimates more than one in five Americans has any mental illness.

Electronics

In this class, we will often do demonstrations of the tasks in the papers we read, so I encourage you to bring your laptop/tablet to class so you can engage with those activities. However, research shows that using electronic devices in class can reduce focus and lead to more superficial learning (for you and those nearby). For that reason, laptops and tablets are allowed for note-taking or class-activities only.

I know I've been guilty of thinking I'll be able to have my laptop out and not be distracted, just to find myself doing other work/email/sudoku five minutes later, so I encourage you to think carefully about how you want to take notes for this course and check in on your choice throughout the semester. If you want a paper notebook, I can provide one!

Helpful Resources

Counseling and Mental Health Services: The Harvard Counseling and Mental Health Services offer a variety of services to address any concerns or additional support you may need. You can call their 24/7 help line at 617-495-2042

The Academic Resource Center: The ARC offers a wide variety of services to support students in their academic goals. These include peer tutoring, workshops on learning and study strategies, setting study groups, or planning a senior thesis.

The Mignone Center for Career Success: The Mignone Center empowers our learners to identify, explore, and achieve their career and professional goals, wherever they may lead.

Food Assistance: Boston Food Bank || Cambridge Food Pantry || Additional Food Pantries

Financial Resources: Griffin Financial Aid Office || Significant Medical Expenses || Computer Loans